

A Critical Review of Deep Learning-Based Clustering Methods for Single-Cell RNA Sequencing Data

Architectures, Benchmarks, and Open Challenges

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Abstract

The single cell data clustering procedure plays key role in the data analysis. Because without a properly chosen cluster, other relevant tasks like rare population and disease characterization wouldn't be achievable. Given high dimension, sparsity, and dropout noise nature of single cell RNA counting data, traditional clustering algorithm, such as k-means, hierarchical clustering, Louvain/Leiden may not perform so well. In this paper, we take an effort to survey nine related papers between 2023 and 2026, which apply deep learning techniques to cluster single cell RNA data. These papers are organized in the perspective of method family, including attention-based/self-supervised autoencoders, graph based embedding, topic modeling/feature selection based embedding, and Transformer foundation models, and then compare their structural assumption, performance, evaluation metrics, and evaluate the trade off for different architectures rather than review each one independently. The paper found that graph based methods yielded consistent better improvement against classic methods, transformer foundation model (scGPT as representative) favored multi-task learning and generalization instead of top benchmark accuracy, while the lack of standardization of benchmark evaluation framework, in parallel to CHM13/QUAST metrics in genome assembly, makes it hard to answer which framework class dominates the most or merely fitted the data of the authors the best.

1 Introduction

Single-cell RNA sequencing determines gene expression on a per-cell basis instead of inbulk tissue averaging, exposing cell heterogeneity inaccessible to bulk techniques. Unsupervised clustering-collating together cells with the highest expression similarity-forms the typical beginning of analyses with scRNA-seq data on novel projects since it requires cell type labels, the ground truth, at the beginning of the analysis process. Given that this aspect is fundamental to downstream studies such as rare cell identification and analyses of disease states, cell grouping errors translate into biological errors. Early clustering pipelines typically used simple, classical approaches such as k-means, hierarchical clustering and graph- community-detection methods (such as Louvain, Leiden), normally in conjunction with a linear reduction such as PCA [1,2]. Though cheap and well-understood, these algorithms were not designed to accommodate the intrinsic statistical properties of scRNA-seq count data, the majority of which are often zeros due to dropout, rather than biological lack of expression. Deep-learning approaches entered the field by directly addresssing this discrepancy. Although [3, 4] extend zero-inflated-negative-binomial autoencoders, they focus on different problem regimes; in [3] cluster number is not pre-assigned, and in [4], finding rare cell type is the goal. A further trend has departed from considering each cells as independently described in an expression space, but rather, explicit models the relationships between cells; scCDCG [5], scV-GATAE [6] and scSiameseClu [7], all construct cell-cell similarity graph then employ a type of graph neural network to re- fine the embedding of cells, this line of approaches show the greater than previous methods as shown consistently across these reports. A separate trajectory pursues generalization rather than task-specific accuracy: scGPT [2] is pretrained once on over 33 million cells and applied across clustering, annotation, batch integration, and perturbation-response prediction via fine-tuning or zero-shot transfer, trading a single fixed benchmark score for cross-task transferability. Across all of these methods, evaluation converges on a small set of recurring

metrics – Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and clustering accuracy (ACC) – computed against reference labels that are themselves usually the output of an earlier annotation pipeline, which makes evaluation partly circular in a way the primary literature rarely acknowledges. Although such a flurry of specialized architectures has emerged, there has been no convergence over which architectural philosophy is actually “best.” The paper above each chose its own baselines, datasets, and preprocessing, meaning even when the same public dataset was applied, it is impossible to compare the reported gains between papers. This survey is a response to this situation; the purpose is not to add yet another private summary of a method, but rather to consciously read the nine papers to see if by reading them against one another-comparing structural assumptions, benchmarking decisions and reported results-we can identify something difficult to do if they are read in isolation: that good incremental results are available despite a missing public protocol for understanding such gains. Here, my specific contributions are :

- Arranging nine deep learning based scRNA seq clustering papers (2023-2026) by four method families into a taxonomy table (Table 2) instead of into per-paper review. (Table 2) rather than a paper-by-paper summary.
- Identifying between multiple methods built on the same architecture yet fine-tuned toward failure-modes (like flexible cluster number vs recovering rare populations) that are often posed as competitive solutions.
- Comparing method families based on the amount of reported performance and attempt to correlate this with specific non-standard design choices like graph construction hyperparameters for the graph family.
- Draw an explicit analogy to genome assembly benchmarking (CHM13 reference, QUASt evaluation) to characterize what scRNA-seq clustering evaluation is currently missing.
- Surface open challenges – standardized benchmarking, hidden hyperparameters in graph construction, and the computational accessibility of foundation models – as the field’s central unresolved issues rather than minor caveats.

2 Paper Selection

The papers were identified through Google Scholar, PubMed, and arXiv searches using the terms single-cell RNA-seq clustering, deep clustering scRNA-seq, graph neural network single-cell, scRNA-seq autoencoder, and single-cell foundation model. The reference lists of the resulting papers were then manually screened, leading to the graph-based group completion. The papers were included if their main contribution was a deep learning method that either directly produces a clustering or an embedding followed by a clustering step for scRNA-seq analysis; if they report any quantitative clustering results on a public data set; and if the paper was published in 2023 or later, thus focusing the review on recent advances. The papers only reporting classical pipelines, using clustering as a downstream task, or not providing any numerical evaluation were excluded. In total, there were nine papers, including both peer-reviewed articles and preprints, that met all of the inclusion criteria. Three of the papers cited below are not included in the above list and are mentioned only for context. Peng et al. [1] and Zhang et al. [2] are review papers published earlier to provide the context of classical baselines, against which the methods reviewed in this article are often compared. Baron et al. [9], Shekhar et al. [10], and Zheng et al. [11] are the original single-cell dataset publications, cited when particular datasets are mentioned. One final note pertains to the selection process: reading the topic-model and featuresselection category was still in progress at the time of this writing, which is why it is described in less detail than the other four. The “material” of this review therefore consists of these nine peer-reviewed and preprint papers on deep learning-based single-cell RNA-seq clustering, published in 2023–2026, together with the public bachelor datasets used for benchmarking by the authors. The method consists of taxonomy-driven reading and writing: the papers are grouped according to the core design principles and then compared within and between groups according to three axes: the assumptions encoded in the architecture (what the model knows about cells and their relationships); the claims of performance and limitations made by the authors; and the evaluation protocols used to support the claims. Table 2 below summarizes this taxonomy.

3 Background

An scRNA-seq experiment produces a count matrix whose rows are individual cells and whose columns are genes, for this reason an experiment already has tens of thousands of columns and tens of thousands of rows. Two properties of this matrix shape every method in this review. It is extremely sparse, often with more than ninety percent zeros, and only some of those zeros are biological: many are dropout events, where a transcript was present in the cell but never captured during library preparation. As a zero might signal "not detected," or "not detected with a slight, but not insignificantly slight, expression that can be treated as absent," distance metrics treating the matrix as just arbitrary numbers perform poorly, and for this reason, various kinds of the models whose performance was assessed, that assign a ZINB (zero-inflated negative binomial) likelihood to the reconstruction, which is not simply to the square error. There are just four necessary ideas to digest all nine studies. An autoencoder, taking in a cell, compresses the cells into a short vector (whose members capture the variation in the cells) which is then used to reconstruct the number of readings in the original cell-the latent space is effectively a denoised version, ready to be clustered. An attention layer enables a particular portion of the input to be given a greater or lesser value than others, rather than all features of the input having the same role. Finally, a graph neural net builds a representation of the cells with nodes as cells, a graph of nodes based on knearest neighbours or so, or in general, which edges connect similar cells, a cell's representation is updated using neighbour representations, thus stopping cells from sitting alone in their latent representation. A transformer foundation model is pretrained once on a very large cell collection, then adapted by fine-tuning or used directly without further training. Evaluation in this area is almost always external: a clustering is compared against a set of reference cell-type labels. ARI measures agreement between the predicted partition and the reference partition while correcting for agreement expected by chance, NMI measures how much information the two partitions share, and ACC reports the fraction of cells assigned to the correct cluster after the best matching between predicted and reference clusters. When no labels exist, the silhouette score is sometimes used instead, which only measures how compact and well separated the clusters are. Table 1 lists the abbreviations used from here on.

4 Comparative Review of Existing Methods

4.1 Methods and Algorithms

The nine reviewed papers fall into the four families introduced in Table 1.

Table 1: Taxonomy of the four methodological families identified across the nine reviewed papers, with representative works per branch.

Deep learning approaches to scRNA-seq clustering / embedding		
Family	Defining mechanism	Representative works
Attention / self-supervised autoencoder	Denoising autoencoder with attention; each cell encoded independently	Li et al. 2023 [4]; Lei et al. 2023 [3]
Graph-based (GNN / GAT / graph embedding)	Cell-cell similarity graph refined by a graph neural network	Xu et al. 2024 [8]; Liu et al. 2024 [5]; Xu et al. 2025 [9]
Topic-model / feature-selection embedding	Interpretable embedding via topic structure or sparse gene selection	scE2TM; FSSC (in progress)
Transformer foundation model	Large-scale pretraining, then fine-tuning or zero-shot transfer	Cui et al. 2024 (scGPT) [2]

Attention-based and self-supervised autoencoders. Li et al. [4] (AttentionAE-sc) and Lei et al. [3] (DeepScena) both build on the zero-inflated-negative-binomial autoencoder tradition. AttentionAE-sc adds an attention mechanism that fuses denoising with cluster-friendly representation learning, with the main advantage that

Table 2: Taxonomy of the four methodological families identified across the nine reviewed papers, with representative works per branch.

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it does not require the number of clusters to be specified beforehand. DeepScena instead targets a narrower but practically important failure mode: the tendency of earlier deep clustering methods to merge or overlook rare cell subpopulations, and is evaluated specifically on recovering rare populations such as subpopulations within CD14 monocytes in a multiple myeloma dataset.

Graph-based embedding methods. scCDCG [8], scVGATAE [5], and scSiameseClu [9] each explicitly build a graph in which cells are nodes and edges encode similarity, then refine embeddings with a graph neural network variant – graph-cut-informed embedding, graph attention, and Siamese dual-augmentation with optimal-transport clustering, respectively.

Topic-model / feature-selection embedding. scE2TM and FSSC build embeddings intended to be directly interpretable, via topic-model structure or explicit sparse gene selection. Detailed reading of this family was still in progress at the time of writing (see Table 2).

Transformer foundation models. scGPT [2] is pretrained once on over 33 million cells and then applied to many downstream tasks – clustering, cell-type annotation, batch integration, and perturbation-response prediction – through fine-tuning or zero-shot transfer, mirroring a broader shift also visible in genome assembly, where general-purpose long-read assemblers replaced pipelines built for one species at a time.

4.2 Datasets and Biological Setting

The papers reviewed here do not share a single dataset, but a small number of public scRNA-seq datasets recur across them and function as de facto community benchmarks. Table 3 summarizes the datasets referenced across the reviewed literature, rather than data collected for this review itself. Unlike genome assembly, where a near-complete reference genome (the T2T-CHM13 human assembly) provides an objective ground truth, scRNA-seq clustering has no real equivalent: the “ground-truth” cell-type labels used to compute ARI or NMI in these datasets are themselves usually the output of an earlier annotation pipeline.

The recurrence of PBMC-derived and pancreas datasets across papers from otherwise unrelated research groups suggests an informal convergence on a benchmark set. However, unlike QUASt-style assembly benchmarking against CHM13, there is no agreed protocol for preprocessing, gene filtering, or metric computation, so reported numbers cannot be compared across papers as directly as they might first appear (see Section 4.3). The biological settings are also less interchangeable than the shared metric names imply. PBMC data contains many well separated immune populations and a few closely related ones, the Baron pancreas islets contain cell types of very unequal abundance, the Goolam embryo dataset has only a small number of cells, and the Shekhar retina data contains fine-grained bipolar subtypes. A method that performs well on a small, clean embryo dataset is therefore not being asked the same question as one tested on tens of thousands of immune cells.

Table 3: Datasets referenced across the reviewed papers.

Dataset	Details	Used in
PBMC (Zheng 68k)	10x Genomics; ~68,000 human PBMCs [11]	Common benchmark across nearly all papers reviewed
Baron pancreas (human & mouse)	Islet cell types [1]	scVGATAE
Goolam mouse embryo	Early mouse embryonic development, small cell count	scVGATAE
Human cerebellum	Human cerebellar cell atlas	scVGATAE
Shekhar mouse retina	~27,000 bipolar retina cells [7]	scSiameseClu
Human liver cells	Human liver cell atlas	scSiameseClu

4.3 Evaluation Strategies

Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and clustering accuracy (ACC) dominate the evaluation protocols across all reviewed papers, occasionally supplemented by silhouette score in label-free settings. Because this is a comparative review rather than an original implementation, no train/validation/test split was performed by this work; instead, the evaluation strategy compared here is the one reported by each original paper, and Section 6 assesses how comparable those self-reported protocols actually are across papers. This review does not implement or retrain any of the nine methods; the analysis in this section compares the results as reported in the original papers rather than results independently reproduced by this work. Table 4 summarizes the clearest quantitative claims found across the reviewed literature. What the two rows of that table have in common is more revealing than the numbers themselves: in both cases the comparison basis is a baseline set chosen by the authors of the method being promoted, on a dataset those same authors selected. Neither figure is wrong, but neither was produced under a protocol that anyone outside the paper controlled.

Table 4: Selected quantitative results as reported in the original papers.

Paper	Reported result	Comparison basis
scVGATAE [5]	23.2% relative ARI improvement	Best of the paper’s own baselines, Goolam dataset
scSiameseClu [9]	89.16% clustering accuracy vs. 79.84%	Strongest of nine baselines, Shekhar retina dataset

5 Preprocessing and Experimental Settings

For each method introduced here, a single object as input and a mostly same precursor set appear in sequence before any learning step starts. Very low-gene or very few-cell genes are excluded early, since both tend to increase noise. Per-cell counts are normalized against sequencing depth to treat two cells of same state sequenced to different depths equivalently, then normalized counts are log-transformed in order to prevent a few high-gene cells from swamping distance calculations for all cells, and some pipelines furthermore remove the bulk of genes (that vary least and thus cannot help differentiate cell types).

Scaling of each gene by unit-variance may precede analysis or not, depending on the model. Data is drawn from the same data sets.

The four families proceed to do with the matrix before training differ, though it may not be obvious how. The ZINB autoencoder approaches must take along raw cell-by-gene counts because they define the reconstruction error (that used in the loss function) not on normalized counts but on raw count probabilities of some count generated (after normalization and encoding and passing the latent space) that the decoder is trained to mimic. The graph-based methods require additionally a graph, built based upon cell-cell similarities (calculated from expression data after PCA, etc.), so their input is a graph-based object.

Graph formation usually hinges on a k-nearest neighbor calculation, with decision of neighbor number, distance metric, and weighting of edges being part of protocol.

Transformer models use gene IDs and expressions (often quantized to bins to form discrete tokens) such that their inputs become sequences, allowing them to apply one learned model to multiple datasets.

The setup under which this survey operates must also be specified, as it is not one of an implementation paper. No models are trained here, hence there is no test set/training set split, no fixed seeds, no epoch budgets or learning rates and no details regarding hardware. All data were generated elsewhere, as described herein. Each analysis thus represents a reported result from one research group in a report based upon reported results from various different groups, compared in terms of which protocol best achieves the best reported results using metrics, and not of an experimental result generated in the course of this paper.

The issue is compounded by varying levels of detail in published protocols, with less consistency in how the choices above are made, and particularly variable consistency on graph-neighbor counts in that approach and number of variables to include of both. Baselines likewise vary widely between studies (an old, well-performing baseline cannot equal a more new one in the same ways that it cannot match state-of-art). Reported runtime varies wildly.

Mostly, transformer models often omit details critical to ordinary lab computer analysis.

These issues, however, don't prove specific research outcomes incorrect. Rather, they can lead to reported ARI scores for the same data sets which reflect two different quantities, so this analysis takes shared pre-training as necessary for fair comparison.

6 Cross-Paper Comparison

Here, table 6 contains the nine papers horizontally across those parameters that seem relevant to this review: namely, the problem, the mechanism, the datasets, and the strengths and the limitations.

Reading across rows, the table shows a field moving in one direction on structure and in two directions on purpose. Every family after 2023 adds relational or contextual information that the plain autoencoder did not have, whether through a graph or through pretraining. What they add it for differs: the graph-based papers spend it on accuracy against reference labels, scGPT spends it on transfer across tasks, and the topic-model family spends it on interpretability. Because these are three different objectives, the highest ARI in the table is not automatically the best method, and PBMC is the only column value shared widely enough to support even an informal comparison.

7 Discussion

Two limitations recur across the field and directly bound how much can be concluded from Table 5. First, as discussed in Section 4.3, the reference labels ARI, NMI, and ACC are computed against are themselves usually the product of earlier, imperfect annotation pipelines rather than an independent ground truth – a circularity broadly similar to the problem QUASt-based evaluation faces in genome assembly when the reference assembly used as “truth” is itself imperfect. Second, none of the gains summarized in Table 5 have been independently reproduced on a common benchmark suite; each paper selects its own baselines and dataset combination, which makes cross-paper comparison suggestive rather than rigorous. A genuine ablation study – isolating, for example, the effect of graph-construction choices (k-NN size, similarity metric) from the rest of the graph-based architectures in Section 4.1 – was not conducted as part of this work, since doing so would require retraining these models rather than reviewing their reported results. This is flagged here explicitly, rather than omitted, as the most direct way this review's scope differs from an implementation-based project, and is consistent with Table 3's observation that graph construction remains a hidden, non-standardized hyperparameter across the literature. Beyond those two limits, three trends hold across the nine papers. The clearest is the shift from modelling cells independently

Table 5: A synopsis of the literature examined.

Method (year)	Problem addressed	Method / main focus	Datasets used	Strengths	Limitations
AttentionAE-sc (2023) [4]	Cluster count must normally be fixed in advance	Attention-based ZINB autoencoder joining denoising with clustering	PBMC and other public benchmarks	Attention-based clustering without predefined cluster number	Cells encoded independently; no cell-cell structure
DeepScena (2023) [3]	Rare subpopulations merged or missed	Self-supervised hierarchical deep clustering	Multiple myeloma (CD14 monocytes), PBMC	Rare cell population detection	Narrow evaluation focus; no relational structure
scCDCG (2024) [8]	Expression-only models ignore cell relationships	Deep cut-informed graph embedding	PBMC and other public benchmarks	Graph-based cell representation learning	Graph construction left as a hidden hyperparameter
scVGATAE (2024) [5]	Fixed-weight neighbourhoods treat all neighbours alike	Variational graph attention autoencoder	Baron pancreas, Goolam, human cerebellum, PBMC	23.2% relative ARI gain on Goolam	Gain measured only against the authors' own baselines
scGPT (2024) [2]	Task-specific models do not transfer	Transformer foundation model for multiple downstream tasks; >33M cells pretraining	Large pretraining corpus plus downstream task datasets	Serves clustering, annotation, integration and perturbation prediction	Pretraining cost beyond typical academic compute
scSiameseClu (2025) [9]	Embeddings unstable and hard to interpret	Siamese graph-based clustering; dual-augmentation with optimal transport	Shekhar retina, human liver, PBMC	89.16% ACC vs. 79.84% for the best of nine baselines	Baseline set again chosen by the authors
scE2TM (2025)	Latent dimensions are not interpretable	Topic-model-based interpretable embedding	Not recorded; reading in progress	Reported gain in interpretability	Not yet fully integrated into this comparison
FSSC (2026)	Too many genes contribute noise	Sparse feature-selection embedding	Not recorded; reading in progress	Explicit gene selection aids interpretation	Reading in progress at time of writing

to modelling them relationally, and it is also the shift with the largest reported payoff. The second is a move from single-purpose models toward general ones, which scGPT represents on its own and which is measured differently from everything else in this review. The third is a growing concern with interpretability, visible in the topic-model and feature-selection family, which accepts that a slightly worse partition may be preferable if a biologist can say why the partition came out that way. The papers also disagree, though the disagreement is mostly implicit. AttentionAE-sc treats having to fix the cluster number as the central weakness of deep clustering, while DeepScena accepts that constraint and treats rare-population recovery as the real problem, even though both sit on the same ZINB autoencoder foundation. Reading them as rivals is a mistake: they solve different failures of a shared design. Similarly, the graphbased papers disagree only about how to weight neighbours, not about whether the neighbourhood graph should exist. What everyone agrees on is the metric set, which is unfortunate, because agreement on metrics without agreement on preprocessing produces numbers that look comparable and are not.

8 Research Gaps and Future Directions

Three concrete gaps follow from this: (1) the field lacks a shared benchmarking framework comparable to CHM13/QUAST in genome assembly; (2) graph-construction choices in the strongest-performing family are a hidden hyperparameter that has not been isolated via ablation; and (3) it remains unclear how much of scGPT's benefit survives when fine-tuned on modest, lab-scale data rather than used as a pretrained embedding extractor. Extending these architectures to multi-omic and spatial transcriptomics data, and to underrepresented tissues and rare cell types not well covered by current pretraining corpora, remains a natural direction for future work. Each gap suggests something realistic to do next. The first needs a fixed benchmark suite rather than a new model: an agreed list of datasets, one published preprocessing script, and a rule that every method reports on all of them,

so that a gain is a property of the method and not of the pipeline in front of it. The second needs a controlled sweep over k-NN size and similarity metric for one graph-based model, holding everything else fixed, which is cheap compared with proposing another architecture and would show how much of the graphbased advantage is architectural at all. The third needs fine-tuning experiments at the data and compute scale an ordinary lab actually has, reported together with runtime and memory cost, since a model no one can afford to adapt is not usable regardless of its benchmark numbers. A smaller gap is imbalance: rare-population accuracy is claimed by one paper and left largely untested by the rest.

9 Conclusion

Deep learning-based clustering of scRNA-seq data has clearly moved past the point where classical methods are competitive on the benchmarks reported in the reviewed literature, and graph-based methods in particular show substantial, repeated gains over non-graph baselines. Whether this reflects a genuinely superior architectural paradigm, though, or simply an artifact of non-standardized evaluation, is not something the current literature resolves, and this review treats that as the most important open question in the field rather than a minor caveat. The main limitation of this review; as the comparisons here solely rely on the result that was self-reported in the paper without reproduce. The logical and appropriate next step is an implementation test across representative methods within each family on a same dataset under same pre-processing pipeline in a controlled method.

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